

ASCII MASTER FLOW – PANEL 1/12: Data-operation decision board

BIT / BYTE -> quantity

ENCODING -> representation

COMPRESSION -> size

HASHING -> fixed digest

ENCRYPTION -> confidentiality with key

ASCII MASTER FLOW – PANEL 2/12: Complete computing stack

PHYSICAL -> semiconductor + CPU + memory + storage + I/O

SYSTEM -> firmware + OS + driver + file system

APPLICATION -> program + database + service

NETWORK -> protocols + devices + infrastructure

GOVERNANCE -> security + access + energy + e-waste

ASCII MASTER FLOW – PANEL 3/12: CPU and memory hierarchy

FETCH -> DECODE -> EXECUTE -> STORE

REGISTER -> CACHE -> RAM -> SSD/HDD

FAST / SMALL -----> SLOW / LARGE

PERFORMANCE -> workload + IPC + bandwidth + thermal

CLOCK RATE ALONE -> insufficient

ASCII MASTER FLOW – PANEL 4/12: Processor and embedded map

MICROPROCESSOR -> general CPU

MICROCONTROLLER -> CPU + memory + interfaces

SoC -> multiple system components

GPU / ACCELERATOR -> parallel workload

SENSOR -> measure | PROCESSOR -> compute | ACTUATOR -> act

ASCII MASTER FLOW – PANEL 5/12: Program and execution ladder

SOURCE -> compiler / interpreter / assembler

PROGRAM -> PROCESS -> THREAD

CONCURRENCY -> overlapping progress

PARALLELISM -> simultaneous execution

RACE / DEADLOCK -> coordination failures

ASCII MASTER FLOW – PANEL 6/12: Network device and layer map

APPLICATION -> HTTP | DNS

TRANSPORT -> TCP | UDP

INTERNET -> IP + router

LINK -> Ethernet | Wi-Fi + switch

PHYSICAL -> electrical | optical | radio

ASCII MASTER FLOW – PANEL 7/12: Internet and Web request flow

DOMAIN --DNS--> IP ADDRESS

BROWSER --HTTP/HTTPS--> WEB SERVICE

TCP/UDP -> transport | IP -> routing

COOKIE -> browser-held state / identifier

HTTPS -> protected transport, not truthful content

ASCII MASTER FLOW – PANEL 8/12: Performance and wireless matrix

BANDWIDTH -> capacity | THROUGHPUT -> achieved rate

LATENCY -> delay | JITTER -> delay variation

LTE -> packet cellular | VoLTE -> voice over LTE

Wi-Fi | Bluetooth | NFC | RFID | VLC -> distinct roles

WIRELESS LABEL -> no universal range or wall penetration

ASCII MASTER FLOW – PANEL 9/12: Virtualisation and cloud boundary

VM -> virtual hardware + guest OS

CONTAINER -> shared host kernel

ORCHESTRATION -> manages many services

CLOUD -> five NIST characteristics

IaaS | PaaS | SaaS -> responsibility shifts

ASCII MASTER FLOW – PANEL 10/12: Data and distributed trade-off

DATABASE / DBMS -> organised managed data

ACID -> transaction properties

REPLICATION -> availability/locality, not backup

CAP DURING PARTITION -> consistency or availability trade

EDGE / FOG / CLOUD -> placement by latency and resilience

MUST REMEMBER: Computing separates hardware, firmware, operating system, application and...

ASCII MASTER FLOW – PANEL 11/12: Ledger claim firewall

DISTRIBUTED LEDGER -> replicated record

BLOCKCHAIN -> linked blocks + consensus

PUBLIC | PRIVATE | CONSORTIUM -> access models

ORACLE INPUT -> can be false

NFT / WEB3 / IMMUTABILITY -> conditional claims

CLOSE DISTINCTION: Bit is not byte, RAM is not storage, core count is not performance,...

ASCII MASTER FLOW – PANEL 12/12: Advanced computing and status rail

HPC -> processor + accelerator + interconnect + software

AI -> software capability | SEMICONDUCTOR -> device base

QUANTUM -> qubits for selected problems

SECURITY -> confidentiality + integrity + availability

STANDARD -> DEMO -> DEPLOYMENT -> VERIFIED OUTCOME

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State layer, component, protocol,...